

Abhishek Janagoudar

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📄 Visa Valid till Feb 2027



Profile

Master's student in Electrical and Information Technology, specializing in Automation and Robotics at Hochschule Darmstadt. Gained hands-on experience in mobile robotics during an internship at Baer Automation. Actively seeking thesis opportunities in autonomous navigation, computer vision, and robotic perception.

Professional Experience

BÄR Automation GmbH, Mobile Robots Development Intern

11/2025 – Present
Stuttgart, Germany

- **Explored and compared SLAM techniques:** Learned how to configure and use SLAM Toolbox and Google Cartographer in ROS 2. I created maps using different modes (like async and lifelong) to see which approach worked best for our mobile robots.
- **Improved robot localization:** Gained hands-on experience tuning sensor settings. I learned how to adjust LiDAR scan matching and odometry parameters to help the robot figure out its location more accurately and reduce drift over time.
- **Worked with AGV VDA 5050 standards.**
- **Built digital twin tools:** I had the opportunity to learn about Asset Administration Shells (AAS v3.1). I helped create a tool that automatically generates these digital assets and submodels for the ARENA2036 project using official AAS3.1 standards and IDTA Templates for Submodels, making sure all the required technical attributes were included correctly.

J. Ray McDermott Engineering Services Private Ltd

02/2024 – 08/2024
Chennai, India

Associate System Engineer - AVEVA Electrical Administrator V12

- Administered AVEVA Electrical V12 for Ruya EPCIC offshore projects 11 & 13, delivering 50+ electrical datasheets and drawings on time.
- Standardized project workflows that reduced rework hours and improved inter-team coordination.
- Led tool training sessions, reducing internal troubleshooting incidents by 30% within 5 months.

Infosys Ltd.

09/2021 – 02/2024
Mysore, India

Senior System Engineer - AVEVA Engineering V15 Electrical & Instrumentation Administrator

- Managed and optimized AVEVA Engineering 15.x for Electrical and Instrumentation projects, including application architecture, operations, and system maintenance.
- Acted as Scrum facilitator, coordinating sprint reviews, retrospectives, and daily stand-ups, promoting Agile practices and CI/CD integration.

System Engineer:

- Configured libraries and settings, including TAS, I/O cards, wiring rules, instruments, and cable catalogs.
- 50+ User Story deliverables (Electrical Load list, Cable Schedule, Reports, Datasheets, and Drawings).
- Developed data model with classes, attributes, associations, LOVs database views, and Kraftwerk Kennzeichen System naming conventions.
- Mentored 8 engineers through recurring technical sessions and supported project builds, deployments, and application setups.

Arkay Energy Ltd.

07/2019 – 07/2019
Ramanathapuram,
India

Electrical Operations and Maintenance Intern

Supported commissioning/testing of substation systems, performed Operations and Maintenance tasks, learned industrial safety.

LARSEN AND TOUBRO ELECTRICALS

06/2019 – 06/2019
Khargone, India

Electrical Commissioning Intern

Assisted in electrical system setup/testing, troubleshooting, and system optimization.

Education

Master of Science (M.Sc) | Electrical and Information Technology,

Darmstadt University of Applied Sciences

09/2024 – present

Darmstadt,
Germany

Course Overview:

Industrial Robotics, Safety In Industrial Automation, Computer Vision, Industry 4.0 Industrial Internet of Things (IIoT) Adaptive Control, Modelling and Identification, Advanced Programming Techniques, State Space Analysis, Human Machine Interface, Advanced Sensors

Bachelor of Engineering (B.E) | Electrical and Electronics, The National Institute of Engineering

Grade: 8.12 CGPA (German Grade: 1.7)

07/2017 – 09/2021

Mysore, India

Projects

Robotics and Computer Vision Projects

- 3-DOF Robotic Arm: Built and programmed a 3-DOF robotic arm in Webots with servo joints, implementing forward and inverse kinematics, and created a Python interface for Arduino-PC communication.
- Point Cloud Segmentation: Developed a pipeline in Python and ROS for indoor mapping using a Hesai XT16 LiDAR on a UniTree quadruped robot, applying basic statistical and machine learning methods for feature extraction.
- Visual Servoing (Igus ReBeL 6DOF): Used OpenCV template-matching to map image errors to robot joint commands for closed-loop visual control.
- UR5/UR10 Kinematics: Built MATLAB/Simulink models to calculate forward and inverse kinematics and analyze manipulator motion under external loads.

Advanced Programming Technique Lab Projects

- BattleShip Game: Developed a console-based Battleship game in C++ to enhance programming skills, focusing on game logic and user interaction.
- Student Database: Developed and designed an optimized database system for university enrollment using C++ (object-oriented programming), ensuring efficient data management and retrieval.
- Reversi Board Game: Implemented a console-based Reversi board game in C++ to strengthen algorithmic thinking and data handling.

Safety in Industrial Automation Lab:

- Programmed Siemens S7-1516F safety PLC systems and ET200SP modules using TIA Portal, implementing critical safety logic and functions such as ESTOP1, TWO_H_EN, SFDOOR, and FDBACK.
- Conducted rigorous discrepancy analyses and black box testing to ensure safety integrity.
- Evaluated safety functions according to EN ISO 13849-1 standards using SISTEMA software, calculating Performance Levels (PLr), verifying compliance, and enhancing system reliability and safety.

Languages

English

IELTS - 7.5 (C1)

Deutsch

A2.2

Hindi

SKILLS

Robotics & Mobile Robots: Forward & inverse kinematics, DH modeling, Gazebo, RViz, SLAM (SLAMToolbox, Cartographer, AMCL), ROS2, AGV navigation, VDA 5050 compliance, Linux, Git, Docker.

Computer Vision: Image Preprocessing, Camera Calibration, Object Detection, Segmentation, Multi-Object Tracking, LiDAR & Point Cloud Processing

Industrial Internet of Things (IIoT): OPC UA, Node-RED, MQTT, IoT Dashboards, IoT Gateway Integration, Digital Twins, Virtual Commissioning

Automation: TIA Portal, Siemens S7 Safety, PLC Programming, SISTEMA

Programming: Python, C++

Soft Skills: Project Management, Agile/Scrum, Team Collaboration, Technical Training, Troubleshooting, Teamwork, CI/CD pipeline,

Other: Safety in the Utility Industry, Operations, Maintenance, and Commissioning

Engineering Tools: AVEVA Engineering v15.x, AVEVA Electrical v12, Jira, Rally, Software testing